

Bony-tongued fishes

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
ORDER	Osteoglossiformes
FAMILIES	5
SPECIES	244

The **bony-tongued fishes** are named after the tooth-studded tongue and palate found in most species. These relatively large fishes—up to 9¾ft (3 m) long—live mainly in freshwater in the tropics. This group contains such varied forms as the large Amazonian arapaima, the peculiar elephantnose fish (which has an

extended, downward-curving mouth), the eellike knifefish, and the delicate and ornate freshwater butterflyfish. Most are carnivores, feeding chiefly on invertebrates, but a few species eat detritus and plant material.

Anatomy

Although some species also have teeth in their jaws, all bony-tongued fishes have many small, sharp teeth on both their tongue and the roof of their mouth, which are used to exert the main biting motion when seizing prey. Most species in this group have a relatively elongated body with prominent eyes and large, hard scales, often with fine ornamentation, although some have reduced scales. The dorsal and anal fins are usually placed far back on the body and are often very long. Some fishes in this group, particularly the elephantnose

fish and the knifefish, can produce and detect electrical signals, which they use for navigation and communication. In some species, such as the arapaima

and the freshwater butterflyfish, the swim bladder has a lunglike lining so that it can be used as a temporary lung in oxygen-poor water.



ELECTRIC ELEPHANTNOSE
Elephantnose fishes navigate using electrical signals. An electrical field is created around the fish's body by modified muscles, and disturbances in this field are detected by receptor cells.

Chitala chitala

Clown knifefish



Length Up to 34 in (87 cm)
Weight Not recorded
Sex Male/Female
Status Near threatened

Location S. Asia



Pantodon buchholzi

Freshwater butterflyfish



Length 4–5 in (10–13 cm)
Weight Up to 18 oz (500 g)
Sex Male/Female
Status Least concern

Location C. Africa



Arapaima gigas

Arapaima



Length 8¼ft (2.5 m), max 15 ft (4.5 m)
Weight Up to 440 lb (200 kg)
Sex Male/Female
Status Data deficient

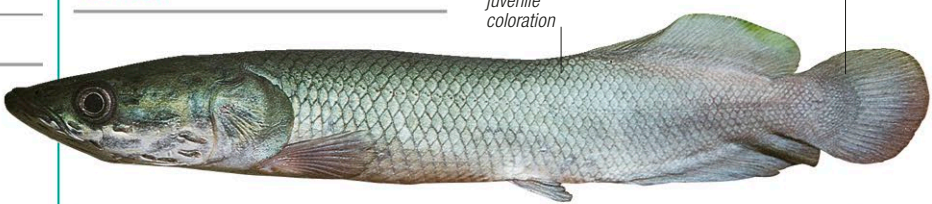
Location N. South America



One of the largest purely freshwater fishes in the world, the arapaima has a streamlined, gray to dark greenish yellow body with long dorsal and anal fins set back close to its tail, a common feature among fishes that make sudden lunges to catch their prey. Apparently unable to fulfil its respiratory requirements with its gills alone, it takes in air from the surface of the water.

light gray-green juvenile coloration

powerful, rounded tail fin



Gnathosomus petersii

Elephantnose fish



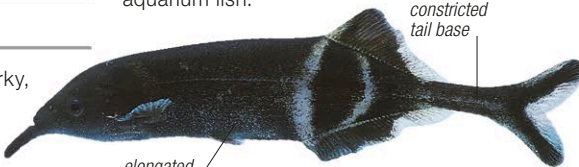
Length Up to 9 in (23 cm)
Weight Up to 2¼ lb (1 kg)
Sex Male/Female
Status Least concern

Location W. to C. Africa



radar to detect obstacles, food, and mates. The highly mobile, fingerlike appendage on its chin is used for probing the muddy bottom in search of food. The elephantnose has an unusually large brain that, in relation to body mass, is equalled only by that of humans. Perhaps for this reason, it has a remarkable ability to learn, and its playful behavior makes it a popular aquarium fish.

constricted tail base

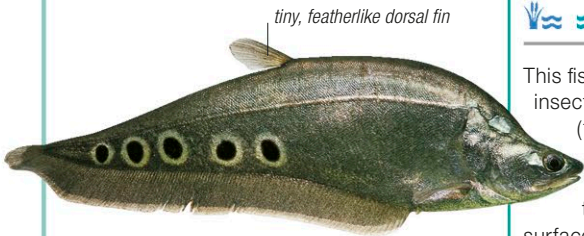
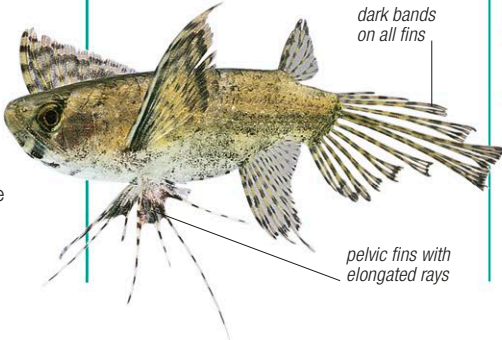


elongated, compressed body

This fish is a surface-feeding insect-eater, able to leap 3¼ft (1 m) or more out of water to catch its prey or escape from predators. In addition to hanging motionless at the surface, it is often seen standing for long periods in shallow water, using its pelvic fin rays as stilts. It has a large, lunglike swim bladder, believed to function as an additional air-breathing organ.

dark bands on all fins

pelvic fins with elongated rays



tiny, featherlike dorsal fin

With a body flattened from side to side, adult clown knifefishes are conspicuously hump-backed, whereas the young are relatively slender. The unusually small dorsal fin is featherlike, giving rise to this fish's other common name, the featherback. This fish swims equally well forward or backward, with an undulating motion of its large anal fin—the main propulsive organ. It has a swim bladder modified to form an accessory breathing organ, and frequently rises to the surface to obtain air. It is often seen at the surface, splashing as it rolls over, exposing its silvery flanks.